

Winter Bootcamp Syllabus

Mathematical Foundations for Finance and Risk Engineering

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The bootcamp will take place during the weeks of January 5 and January 12, running **from Monday, January 5 through Friday, January 16**. Sessions will be held on **Monday, Wednesday, and Friday from 8:30 a.m. to 12:15 p.m. .**

The purpose of this Bootcamp is to provide you with the mathematical tools necessary to handle most of the quantitative concepts you will encounter during the program. The topics covered in this Bootcamp are: **Matrices, Differential Calculus, Optimization, and Probability.**

Except for the first session, which will be entirely dedicated to matrices, each lecture will consist of **1 hour and 30 minutes** of exercise review (based on assignments given from one class to the next), followed by a **15-minute break**, and then a **2-hour lecture**.

All slides and exercises will be made available online. **Standard exercises** will be corrected during class, and I will upload **selected elements of correction** for the advanced exercises.

There will be **no final exam** for this refresher course, nor any **graded homework assignments**. However, it is **highly recommended** that you complete the *standard exercises* between sessions, as they will help you quickly assimilate the concepts studied during the lectures.

If you have any questions, feel free to contact me by email at: **ns6982@nyu.edu**.

1 Matrices (Lecture 1)

The first part of the Bootcamp focuses on the fundamental operations involving matrices and their applications to finance. We will review how to manipulate matrices, solve linear systems, diagonalize matrices and see how it can be used, either to understand the role of covariance matrices in portfolio theory, solve an ODE linked with interest rate models.

Main concepts:

- Definition of matrices and operations on matrices

- Transpose (and symmetric matrices), determinant, and inverse.
- Linear systems: solving $Ax = b$ and interpretation.
- Eigenvalues, eigenvectors, and diagonalization (intuitive approach) including spectral theorem.
- Covariance and correlation matrices with applications.

Applications:

- Covariance and correlation matrices with applications, introduction to Portfolio variance: $\text{Var}(R_p) = w^\top \Sigma w$ and minimum variance portfolios: solving $\Sigma w = \lambda \mathbf{1}$.
- Risk factor decomposition and Principal Component Analysis (PCA).
- Matrix exponential e^{At} and applications to dynamic systems (interest rate model, Vasicek model).

Suggested references to go further

- Gilbert Strang, *Introduction to Linear Algebra*.
- David C. Lay, *Linear Algebra and Its Applications*.
- J. Douglas Carroll and Paul E. Green *Mathematical Tools for Applied Multivariate Analysis*.

2 Differential Calculus (Lecture 2)

This section reviews single and multivariate differential calculus and emphasizes how these concepts appear in optimization, pricing models, and sensitivity analysis.

Main concepts:

- Derivatives and partial derivatives.
- Integration and derivation of integrals
- Gradient and Hessian matrices.
- Taylor expansion and approximation.
- Differentiability of vector-valued functions.

Applications:

- Heat equation, linear-quadratic HJB, 1 dimensional transport equation and their solutions.
- Option pricing and greeks.
- Identifying convexity via the Hessian.

Suggested references:

- Carl P. Simon and Lawrence Blume, *Mathematics for Economists*.
- Sydsaeter and Hammond, *Essential Mathematics for Economic Analysis*.
- J. Douglas Carroll and Paul E. Green *Mathematical Tools for Applied Multivariate Analysis*.

3 Optimization (Lecture 2)

Optimization plays a central role in financial decision-making, from portfolio selection to risk minimization. This part covers both unconstrained and constrained optimization problems and introduces Lagrange multipliers.

Main concepts:

- First- and second-order conditions for optimality.
- Convex and concave functions.
- Lagrange multipliers and constraint qualification.

Applications:

- Mean-variance optimization (Markowitz portfolio theory).
- Budget constraints and shadow prices.
- Calibration problems and least-squares estimation.
- Linear Regression

Suggested references:

- Boyd and Vandenberghe, *Convex Optimization*.
- J. Douglas Carroll and Paul E. Green *Mathematical Tools for Applied Multivariate Analysis*.

4 Probability

This final module of the Bootcamp is devoted to the fundamental concepts of probability theory and their direct applications. The goal is to understand how uncertainty is modeled mathematically and how probabilistic tools support portfolio analysis, pricing, and risk measurement.

4.1 Basics (Lecture 3)

We begin with the foundations of probability:

- Definition of a probability space and events.
- Probability rules, independence, and conditional probability.
- Law of total probability and Bayes' theorem.

Applications include modeling discrete events such as default risk, and interpreting joint and conditional probabilities in financial contexts.

4.2 Random Variables (Lectures 3 and 4)

We then introduce random variables as representations of uncertain outcomes:

- Discrete and continuous random variables.
- Probability mass and density functions, distribution and cumulative functions.

- Expected value, variance, covariance, and correlation.
- Joint and marginal distributions, independence of random variables.
- The normal and multivariate normal distributions and their importance in finance.

Illustrations include the modeling of asset returns, portfolio variance, and the concept of diversification through covariance matrices.

4.3 Conditional Expectation (Lectures 4 and 5)

Conditional expectation is introduced as a key tool for modeling information and forecasting:

- A quick definition of filtrations and sigma-algebra
- Definition of conditional expectation and its interpretation as a projection.
- Definition of (sub/super) Martingale

4.4 Convergence of Random Variables (Lectures 5 and 6)

Finally, we discuss convergence results that justify the use of statistical estimation and simulation:

- Convergence in probability and almost sure convergence.
- Law of Large Numbers (LLN) and its role in empirical estimation.
- Central Limit Theorem (CLT)
- Monte Carlo simulation: generating random samples, estimating means, variances.

These concepts form the theoretical foundation for simulation-based methods used in modern quantitative finance.

Applications:

- Inferring the law of the underlying asset using Call options prices.
- Markov Chains: modeling credit rating transitions, regime-switching models, and stochastic state dynamics in risk processes.
- Monte Carlo Methods: simulating portfolio returns, estimating option prices.
- Law of Large Numbers and CLT in practice: convergence of simulated estimators, confidence intervals, and statistical validation of models.
- Linear Regression and Least Squares Estimation: parameter estimation, CAPM and multifactor models, calibration of linear pricing relationships.

Suggested references:

- Sheldon Ross, *A First Course in Probability*.
- Robert Hogg & Allen Craig, *Introduction to Mathematical Statistics*.